Zhexiang (Bruce) Li

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RESEARCH INTERESTS

Multimodal learning for health; efficient attention and visual token routing; interpretable molecular and spectrum modeling.

EDUCATION

University of California, Los Angeles Sep 2026 - Jun 2028 (expected)

M.S. in Computer Science

University of Southern California May 2026

B.S. in Computer Science · GPA 3.76 / 4.00 · Viterbi Dean's List

RESEARCH EXPERIENCE

Graduate Researcher | Health Intelligence Lab (HAIL), UCLA Aug 2026 - Present

Advisor: Prof. Yuzhe Yang

- Investigate multimodal foundation models for diabetes, integrating continuous glucose monitoring with insulin, meals, activity, and patient context.
- Built patient-disjoint MetaboNet benchmarking and audit pipelines. Evaluated GluFormer, T1DFormer, and custom predictive-state and Transformer architectures at 30/60/120-minute forecasting horizons.

PUBLICATIONS & MANUSCRIPTS

MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification 2026

Zhexiang Li. Preprint · arXiv:2609.30613

Developed a lesion-supervised router after the vision backbone, with budget curriculum learning, diversity regularization, and attention distillation. Evaluated how much annotated lesion evidence remains after routing on ISIC 2019.

Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval 2026

Manuscript submitted to AAAI 2027

Represented mass-conserving fragment pairs as sparse evidence tokens for spectrum retrieval. Used closest-mass and overlap-hard negatives, alongside evidence-removal tests, to examine whether the exported evidence affects ranking.

Efficient Attention via Pre-Scoring: Prioritizing Informative Keys in Transformers 2025

Zhexiang Li, Haoyu Wang, Yutong Bao, David Woodruff. Under review · arXiv:2505.11040

First-author work advised by Prof. David P. Woodruff (CMU). Proposed selecting informative key intervals before attention computation; implemented Triton kernels and compared with HyperAttention and LevAttention.

Game Theory in Companies 2023

S. Li, Z. Li, Z. Liu. Advances in Economics, Management and Political Sciences, 16(1), 357-366. DOI

Advised by Prof. Ming Gu (UC Berkeley). Led a team modeling game-release decisions as two-player normal-form games and enumerating Nash equilibria with Python and Gambit.

PROFESSIONAL EXPERIENCE

AI Engineer Intern | PMAOS / Next Big Seek
Jul - Aug 2026

- Designed bidirectional user-video interest propagation for short-video recommendation. Separated learned preferences, retrieval scores, and diversity annotations, with session, bridge, and long-term exposure budgets and fatigue controls.
- Developed 11-language audio-visual affect inference with evidence fusion and a fixed-point decision layer that can abstain or report conflicting modalities.
- Validated recommendation logic with 629 tests, strict Mypy across 438 source files, and 34 contract checks; the multimodal pipeline passed 139 Python tests and C build gates for Cortex-M3 and ARM926EJ-S.
- Screened sub-10 mm piezoelectric robot architectures and specified a PZT-steel cantilever experiment for a 9 mm candidate; work concluded at experiment and procurement planning.

Full-Stack Engineer | Terlingua Threads, Remote
Jan - May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, SEO, and maintainable content structures.

Co-Founder & CTO | CrAIg Capital, Los Angeles, CA
2025 - Mar 2026

- Led product and engineering for AI-assisted housing debt collection, including data pipelines and automated workflows.
- Set technical direction, hiring priorities, and delivery standards during startup development.

AI Engineer Intern | Piezolab Medical, Lake Forest, CA
Jun - Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Developed agent workflows, experiment checklists, and acceptance tests for vascular-care research.

Business Development Intern | Hangzhou Haoan Jianke Technology, Hangzhou
Jun - Aug 2024

Gathered requirements for a NetEase CodeWave internal platform, helped recruit engineers, and contributed to program structure and interface planning.

Application Engineer Intern | Synaptics, San Jose, CA
Jun - Aug 2023

Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I²C sensor data as MQTT topics for internal dashboards.

Investment Analyst Intern | GUDU Medical, Remote
Jan 2021 - Present

Supported MedTech deal screening, market analysis, due diligence, financial modeling, and valuation.

SELECTED PROJECTS

USC Drug Discovery Research
Spring 2026

Research with a Ph.D. student in Prof. Yan Liu's group

- Studied SARS-CoV-2 antiviral discovery through reproducible RDKit, Chemprop, and hybrid-model benchmarks across seven Therapeutics Data Commons datasets, using scaffold splits.
- Analyzed descriptor importance and SHAP attributions to examine molecular features associated with model predictions.

Fridge Manager
2026

Next.js · Supabase · howtoeat.food

Built meal and grocery planning around available ingredients, leftovers, recipes, dietary preferences, and cooking time.

AI Doctor WeChat

2026

FastAPI · Knowledge retrieval · WeChat Mini Program

Connected a WeChat Mini Program, a browser workspace, knowledge retrieval, and backend services for clinicians, assistants, students, and patient intake.

AI Business Card Hub

Jan 2026 - Present

iOS · Vision parsing · OCR · Source code

Built photo-to-contact extraction with OCR fallback, multi-photo review, bilingual records, and searchable people and company profiles.

TEACHING

Tutor | Introduction to Machine Learning, UCSC

Spring 2023

Prof. Lise Getoor

- Held office hours and discussion sessions on logistic regression, SVMs, and tree models.
- Co-designed Jupyter Notebook assignments and contributed to the coding autograder.

Assistant Coach | Shanghai Bosidi Education

2014 - 2018

VEX Robotics & Olympiad in Informatics

Supported robot design and algorithmic problem-solving instruction, scrimmages, and events.

HONORS & ACTIVITIES

Viterbi Dean's List | University of Southern California

Fall 2024 - Spring 2026

Academic honors.

Volunteer | TIE Hub Co-Founder Speed Date

Sep 19, 2025

Helped organize venue setup, check-in, participant guidance, timekeeping, and event flow.

USC EVMA

Member

Participated as a member of USC EVMA.

TIE Hub Startup Hackathon

Sep 2025

Participated in a five-day workshop spanning concept development, a working demo, venture design, and pitching.

Noah Business School | Entrepreneurship Program

2025

Studied venture building and financing; participated in Silicon Valley company visits and conversations with founders and investors.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript/TypeScript, SQL

Machine learning: PyTorch, Triton, RDKit, Chemprop; multimodal modeling, efficient attention, recommendation, and molecular prediction

Engineering: FastAPI, Next.js, Supabase, Linux; automated testing, static type checking, and data pipelines

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Updated October 2026 · Full experience archive & application resumes